

Class 7: Machine Learning 1

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Today we will begin our exploration of some “classical” machine learning approaches. We will start with clustering:

Let's first make up some data to cluster where we know what the answer should be.

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rmnorm(1000)
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[6] 0.127678633 -1.023370432 -0.354884425 1.046313824 1.441718798
[11] 0.238008904 1.810484732 1.970770224 -0.060756378 -3.188601964
[16] 2.368692624 0.010453232 0.941734492 1.390743565 -1.940728680
[21] -1.420515749 -2.011914050 0.398357074 -0.859802942 -0.062980150
[26] -0.033040832 -0.735072311 -1.631478494 -0.739936267 -0.507126771
[31] -0.471991834 0.250392419 -0.734906836 0.031955346 -1.508721676
[36] 1.419375943 -0.497258060 0.110629246 -1.072752369 -0.424608617
[41] -1.693193992 -1.184460348 -0.027066997 0.311752031 0.036447688
[46] -0.816660881 -0.305833584 1.019967433 -1.478292243 -0.878566075
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[56] 1.254586948 -0.417396836 0.203220138 0.735832687 -2.454563703
[61] 0.867056696 -0.261112094 0.675919279 1.241476751 0.098650150
[66] -1.036591849 0.248890769 0.542113378 1.573485277 -1.409628800
[71] -2.611343916 -1.302761013 -0.616344968 0.575120264 -1.403073408
[76] 3.352287841 -1.217231410 -0.855700739 -0.390828242 -0.399462277
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[86] -0.017353300 -1.210955143 -0.402082789 1.507529074 -0.443781433
[91] 0.146063332 -0.517547797 0.047592838 -1.211683440 -0.453035462
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[116] 0.205696600 0.610199422 -0.604254216 -1.022193762 0.392822198
[121] -0.864380804 -1.597174472 -0.688559653 -0.631876912 -0.696344321
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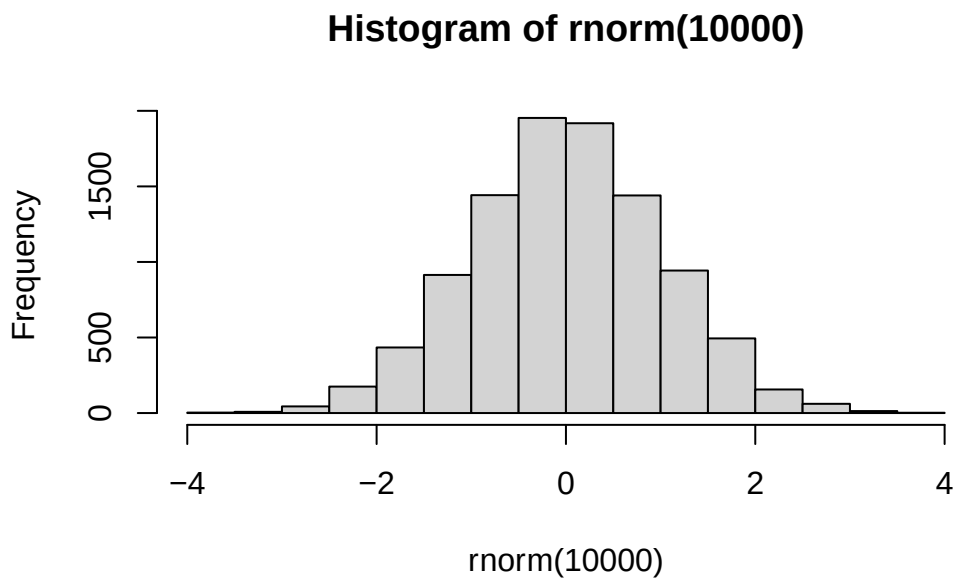
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[956]	-0.714325967	-0.805058283	-0.390347851	0.752954075	-0.021815624
[961]	0.367515993	0.100059179	-1.445205915	-1.763480275	-0.249488073
[966]	-0.242398538	0.213929396	-0.331538535	-0.311857463	-1.340465728
[971]	0.409254494	1.516218697	-1.014728990	1.373718093	1.077024750
[976]	-0.677944649	0.366084277	0.540017493	-2.072179991	0.298217030
[981]	1.109162980	-0.727968130	0.798069798	-0.040273327	-0.452601437

```
[986]  0.733853617  1.143620221  0.625434162  0.903969328  0.705390614
[991] -0.033029426  0.002840082 -0.210141470  0.566840880  0.289033234
[996]  0.098338421  0.031539831 -0.551808643  1.285870912  0.084394599
```

```
hist(rnorm(10000))
```

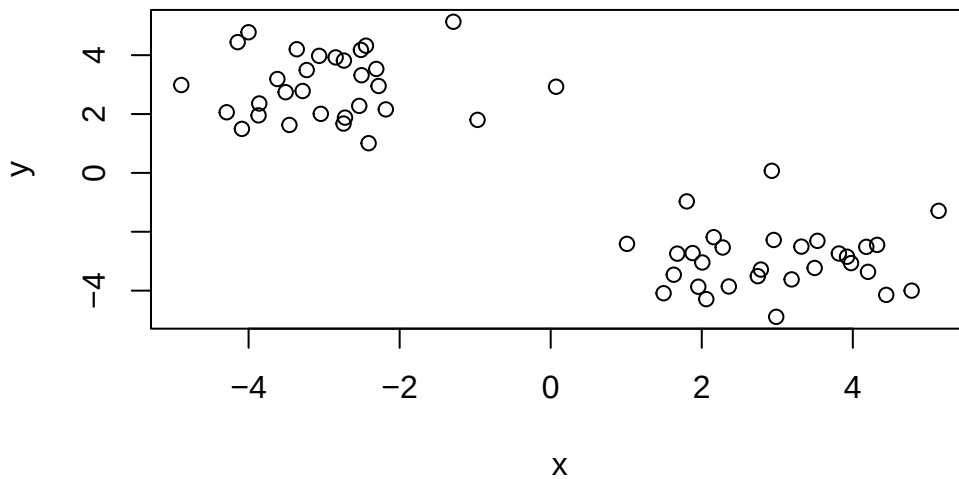


```
x <- c(rnorm(30, mean=-3), rnorm(30, mean = 3))
y <- rev(x)

x <- cbind(x, y)
head(x)
```

```
      x      y
[1,] -2.512596 4.178387
[2,] -3.230294 3.494949
[3,] -2.504455 3.319099
[4,] -3.858411 2.358215
[5,] -2.723239 1.878357
[6,] -4.289356 2.059955
```

```
plot(x)
```



The main function in “base” R for Kmeans clustering is called `kmeans()`.

```
k <- kmeans(x, centers = 2)
k
```

K-means clustering with 2 clusters of sizes 30, 30

Cluster means:

	x	y
1	2.966842	-2.937615
2	-2.937615	2.966842

Clustering vector:

```
[1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1 1
[39] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

Within cluster sum of squares by cluster:

```
[1] 65.74517 65.74517
(between_SS / total_SS = 88.8 %)
```

Available components:

```
[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
```

Q. How big are the clusters (i.e. their size)?

```
k$size
```

```
[1] 30 30
```

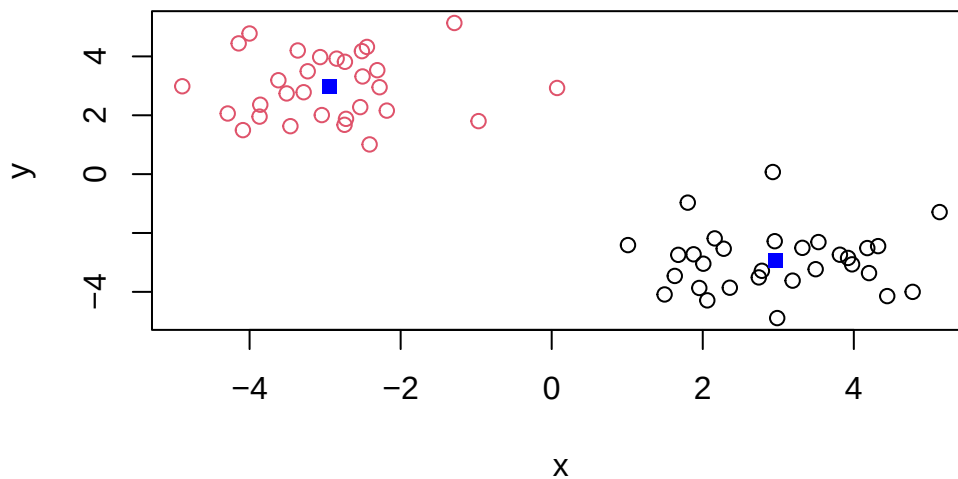
Q. What clusters do my data points reside in?

```
k$cluster
```

```
[1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1
[39] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

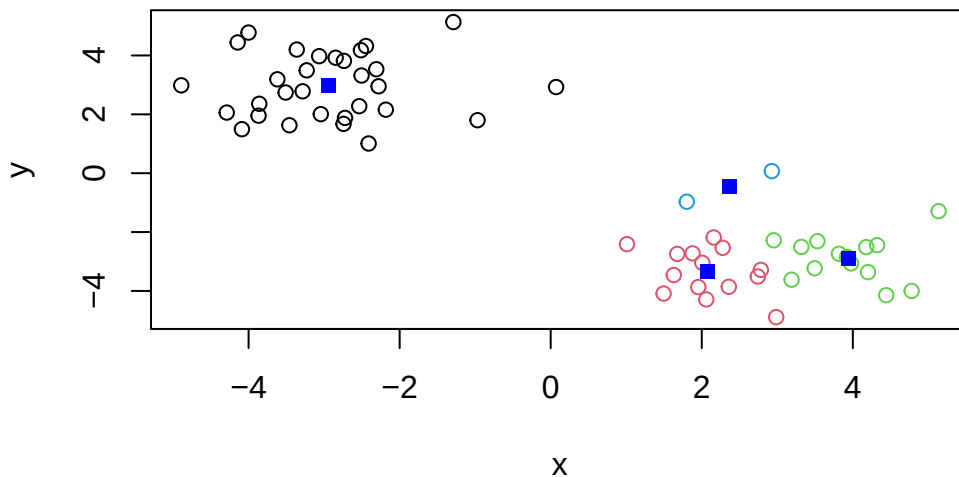
Q. Make a plot of our data colored by cluster assignment - i.e. Make a result figure...

```
plot(x, col=k$cluster)
points(k$centers, col="blue", pch=15)
```



Q. Cluster with k-means into 4 clusters and plot your results as above

```
k4 <- kmeans(x, centers = 4)
plot(x, col=k4$cluster)
points(k4$centers, col="blue", pch=15)
```



Q. Run kmeans with centers (i.e. values of k) equal 1 to 6.

```
k1 <- kmeans(x, centers = 1)$tot.withinss
k2 <- kmeans(x, centers = 2)$tot.withinss
k3 <- kmeans(x, centers = 3)$tot.withinss
k4 <- kmeans(x, centers = 4)$tot.withinss
k5 <- kmeans(x, centers = 5)$tot.withinss
k6 <- kmeans(x, centers = 6)$tot.withinss

ans <- c(k1, k2, k3, k4, k5, k6)
```

Use for loop

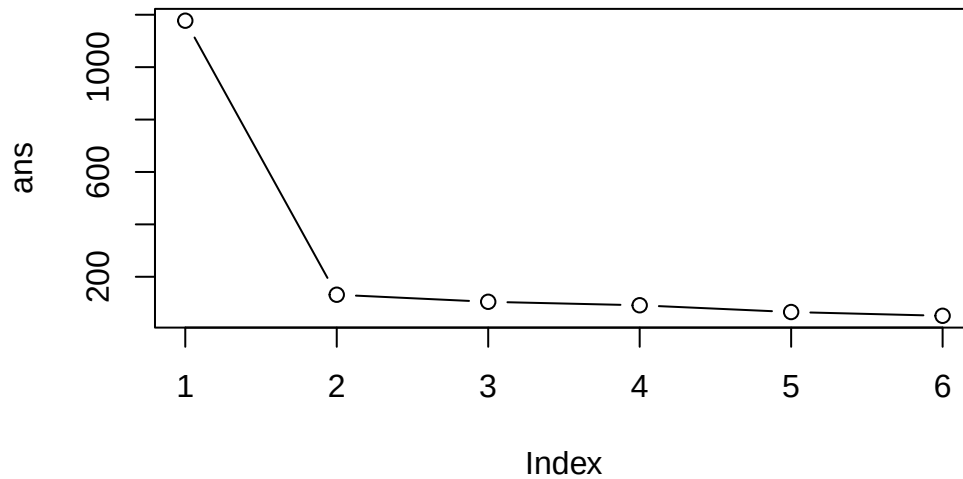
```
ans <- NULL
for (i in 1:6) {
  ans <- c(ans, kmeans(x, centers = i)$tot.withinss)
```

```
}  
ans
```

```
[1] 1177.36857 131.49034 104.53117 91.29937 65.49087 51.10840
```

Make a “scree-plot”

```
plot(ans, type = "b")
```



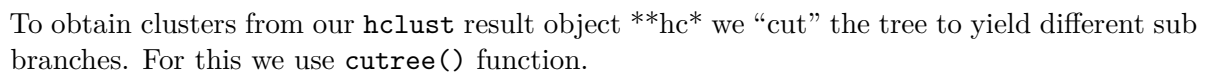
Hierarchical Clustering

The main function in “base” R for this is called `hclust()`

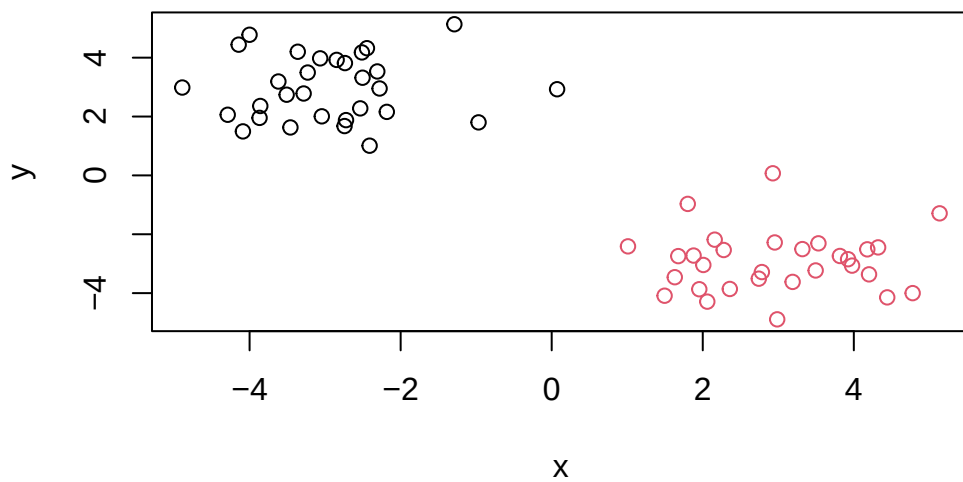
```
d <- dist(x)  
hc <- hclust(d)  
hc
```

Call:
`hclust(d = d)`

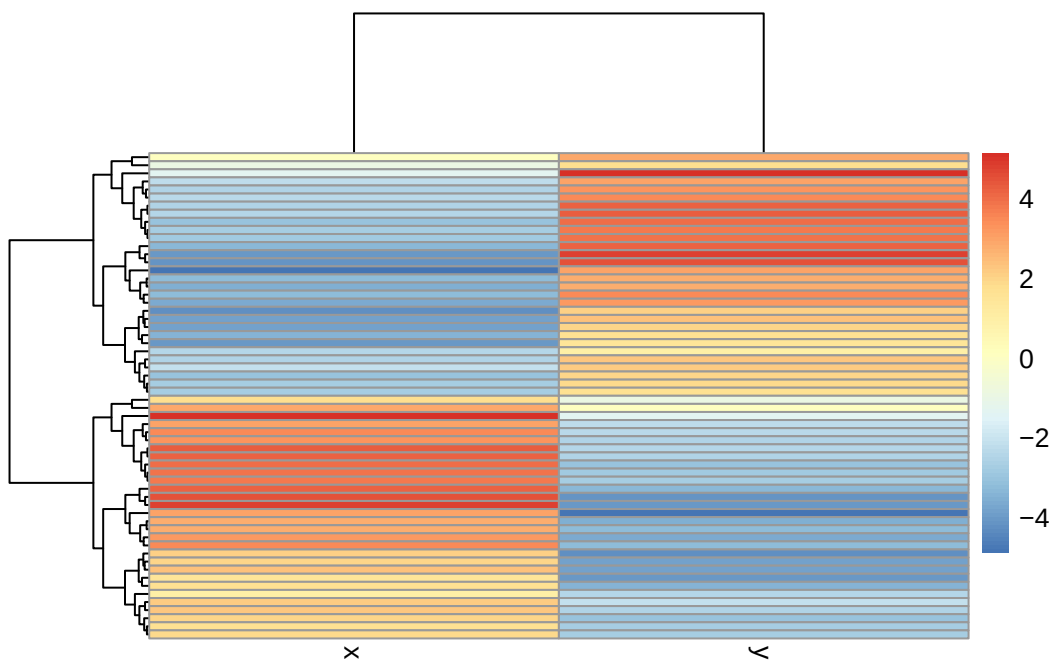
```
plot(hc)
abline(h=7, col="red")
```

[illegible]

11



```
library(pheatmap)
pheatmap(x)
```



Principal Component Analysis (PCA)

```
url <- "https://tinyurl.com/UK-foods"
x <- read.csv(url)
```

Q1. How many rows and columns are in your new data frame named x? What R functions could you use to answer this questions?

```
dim(x)
```

```
[1] 17  5
```

```
rownames(x) <- x[,1]
x <- x[,-1]
```

```
x <- read.csv(url, row.names=1)
head(x)
```

	England	Wales	Scotland	N.Ireland
Cheese	105	103	103	66
Carcass_meat	245	227	242	267
Other_meat	685	803	750	586
Fish	147	160	122	93
Fats_and_oils	193	235	184	209
Sugars	156	175	147	139

Q2. Which approach to solving the 'row-names problem' mentioned above do you prefer and why? Is one approach more robust than another under certain circumstances?

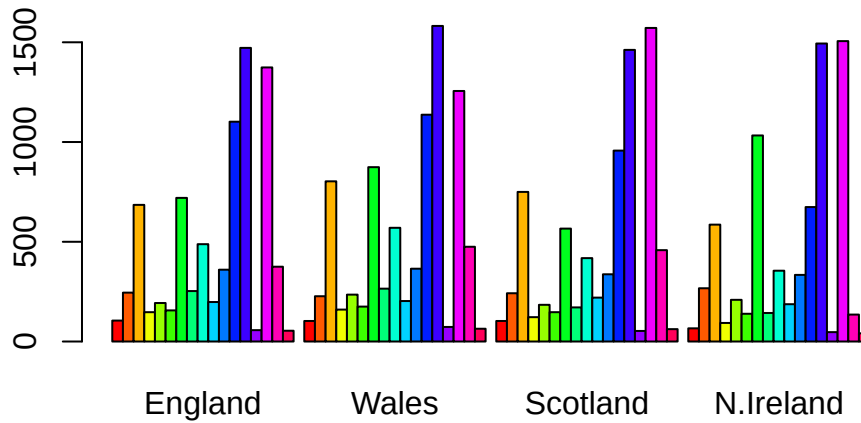
I like fixing it up from when reading my data

Spotting major differences and trends

```
rainbow(nrow(x))
```

```
[1] "#FF0000" "#FF5A00" "#FFB400" "#F0FF00" "#96FF00" "#3CFF00" "#00FF1E"
[8] "#00FF78" "#00FFD2" "#00D2FF" "#0078FF" "#001EFF" "#3C00FF" "#9600FF"
[15] "#F000FF" "#FF00B4" "#FF005A"
```

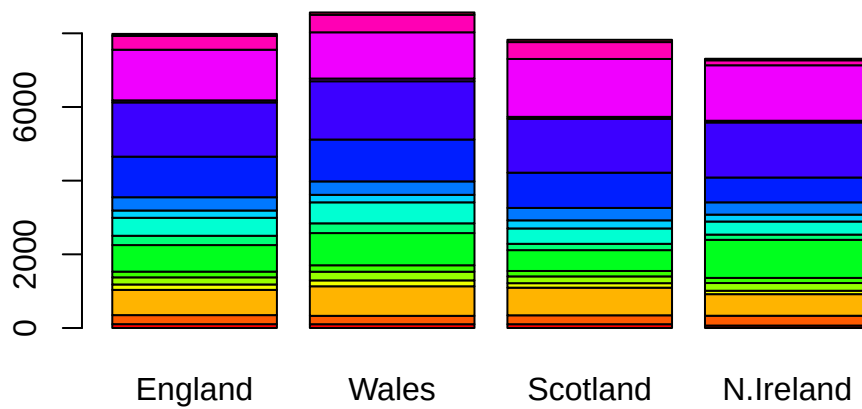
```
barplot(as.matrix(x), beside=T, col=rainbow(nrow(x)))
```



```
rainbow(nrow(x))
```

```
[1] "#FF0000" "#FF5A00" "#FFB400" "#F0FF00" "#96FF00" "#3CFF00" "#00FF1E"
[8] "#00FF78" "#00FFD2" "#00D2FF" "#0078FF" "#001EFF" "#3C00FF" "#9600FF"
[15] "#F000FF" "#FF00B4" "#FF005A"
```

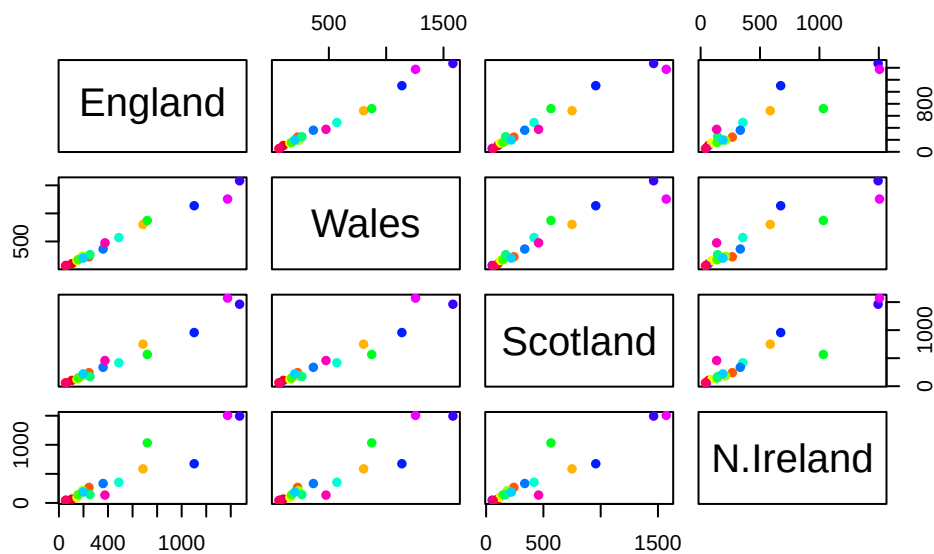
```
barplot(as.matrix(x), beside=F, col=rainbow(nrow(x)))
```



Pair and heatmaps

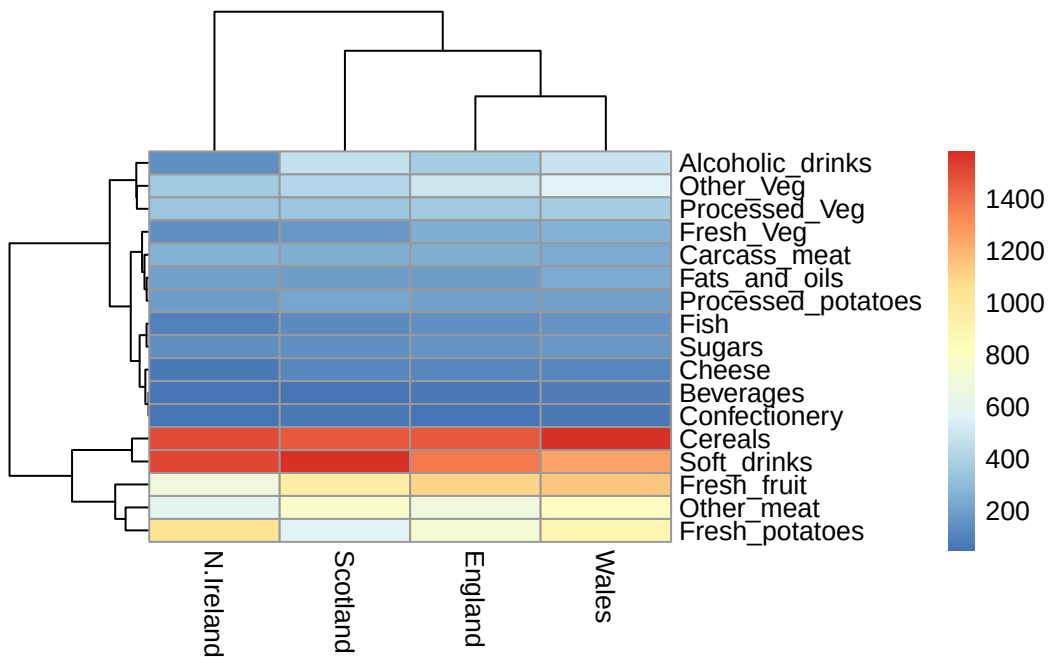
Scatterplot matrices (a.k.a. “pairs plots”) can be useful for relatively small datasets like this one. Let’s have a look:

```
pairs(x, col=rainbow(nrow(x)), pch=16)
```



```
library(pheatmap)

pheatmap( as.matrix(x) )
```



Q6. Based on the pairs and heatmap figures, which countries cluster together and what does this suggest about their food consumption patterns? Can you easily tell what the main differences between N. Ireland and the other countries of the UK in terms of this data-set?

Wales and England are similar in consumption of these foods. It is still quite difficult to say what is going on in the dataset.

PCA to the rescue

The main function in “base” R for PCA is called `prcomp()`.

As we want to do PCA on the food data for the different countries we will want the foods in the columns.

```
pca <- prcomp(t(x))
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	324.1502	212.7478	73.87622	2.921e-14
Proportion of Variance	0.6744	0.2905	0.03503	0.000e+00
Cumulative Proportion	0.6744	0.9650	1.00000	1.000e+00

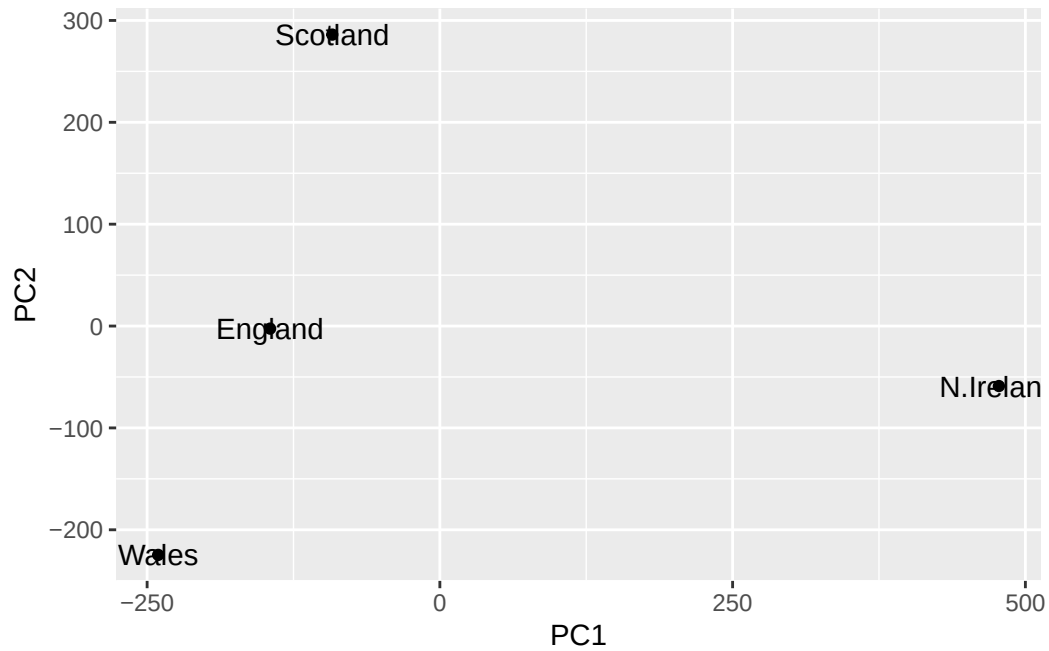
Our result object is called `pca` and it has a `$x` component that we will look at first

```
pca$x
```

	PC1	PC2	PC3	PC4
England	-144.99315	-2.532999	105.768945	-9.152022e-15
Wales	-240.52915	-224.646925	-56.475555	5.560040e-13
Scotland	-91.86934	286.081786	-44.415495	-6.638419e-13
N.Ireland	477.39164	-58.901862	-4.877895	1.329771e-13

```
library(ggplot2)
cols <- c("orange", "red", "blue", "green")

ggplot(pca$x) +
  aes(PC1, PC2, label=rownames(pca$x)) +
  geom_point()+
  geom_text()
```



Another major result out of PCA is the so-called “variable loadings” or `rotation` that tells us how the original variables (foods) contribute to PCs (i.e. our new axis).

```
ggplot(pca$rotation)+  
  aes(PC1, rownames(pca$rotation))+  
  geom_col()
```

